FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

FIITJEE

JEE(Advanced)-2015

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT – IV
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**A, B, C**	**C**	**A, B, C**
2.	**A, B, C, D**	**D**	**A, B**
3.	**B, C**	**C**	**A, C**
4.	**B, C**	**B**	**B, C**
5.	**B, D**	**A**	**A, C, D**
6.	**B, D**	**D**	**A, B, D**
7.	**A, C, D**	**A, B**	**B, C**
8.	**A, B, C**	**A, D**	**A, C**
9.	**C, D**	**B, C**	**B, D**
10.	**A, B**	**A, B**	**A, B, D**
1.	**2**	**6**	**3**
2.	**3**	**5**	**4**
3.	**4**	**0**	**2**
4.	**6**	**6**	**5**
5.	**5**	**3**	**1**
6.	**1**	**3**	**3**
7.	**8**	**1**	**5**
8.	**2**	**2**	**2**
9.	**2**	**5**	**3**
10.	**2**	**5**	**8**

Physics

PART – I

SECTION – A

1. $\int \vec{F}.dx$ is independent of path.

2. A, B, C, D

   Energy received at photo sensitive surface per unit time = $\frac{WS}{4\pi a^2}$

   $\therefore$ number of photons = energy per unit time + energy of a photon.

3. Conservation of momentum and impulse momentum theorem $\int F dt = \Delta P$

4. $\vec{E} = -\left(\frac{\delta v}{\delta x}\hat{i} + \frac{\delta v}{\delta y}\hat{j} + \frac{\delta v}{\delta z}\hat{k}\right)$

6. $\frac{dy}{dx} = 2\cos\left(\frac{\pi x}{L}\right) = \pm 1$

   $\frac{\pi x}{L} = \frac{\pi}{3}, \frac{2\pi}{3}$

   $x = \left(\frac{L}{3}\right), \frac{2L}{3}$

7. $\varepsilon_{ind} = \int \left(\vec{v} \times \vec{B}\right).d\ell$

8. $g_{inside} = \frac{Gm_e}{R^3} r \le R_e$

   $g_{out} = \frac{Gm_e}{r^2}$

   $R \ge R_e$

9. For RLC circuit

   $Z = \sqrt{R^2 + (x_c - x_L)^2}$

   $X_z = \omega L$

   $x_C = \frac{1}{\omega C}$

   Ab = $\omega = \omega'$, $x_L = x_C \Rightarrow z = R$

   Hence phase difference between current and voltage = 0

SECTION – C

1. $f(x,t) = e^{-\left(\frac{ax-bt}{c}\right)^2}$

$v = \frac{b}{A}$

2. $K = \frac{P^2}{2m} = \frac{\left(\frac{h}{\lambda}\right)^2}{2m}$

$K = \frac{h^2}{8L^2m}.9$

3. $R = \int \frac{r^2}{A}\frac{dr}{a\pi r^2} \quad \left|\rho = \frac{1}{\sigma}\right.$

$\frac{(b-a)}{4\pi A}$

4. $u_{cmx} = 5\ m/s$

So impulse = 5m,

$5m.\frac{\ell}{2} = \frac{m\ell^2}{12}\omega$

$\omega = \frac{30}{L}$

5. $\frac{\lambda D}{d} = \beta$

6. $X = h\tan\theta$

$\frac{dx}{dt} = h\sec^2\theta\frac{d\theta}{dt}$

$\frac{d\theta}{dt} = \frac{v}{h\left(1+\tan^2\theta\right)}$

$= \frac{v}{h\left(1+\frac{v^2t^2}{h^2}\right)} = \frac{vh}{\left(h^2+v^2t^2\right)}$

Put $\frac{v}{h} = \omega$

get n = 1

7. $\vec{b} = (q.2a)\hat{z}$

8. $F_3\left(d.v + d.\frac{v}{2}\right)\ dv = (3m + m)g$

$t_{xy} = 0$

$3mg\frac{L}{3} + mgd = dv\frac{L}{2}$

$L\left(mg + \frac{dv}{2}\right) = mgd$

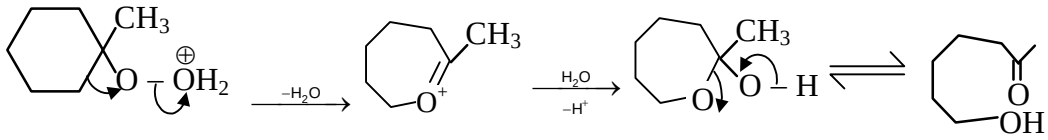
$L = \frac{mgd}{\left(\frac{dv}{2} - mg\right)} = 2d.\frac{(dv - 3mg)}{(dv - 2mg)}$

10. $\lambda = \frac{h}{p} = \frac{h}{eBt} \Rightarrow \frac{d\lambda}{dt} = -\frac{h}{eE}t^{-1}$

Chemistry

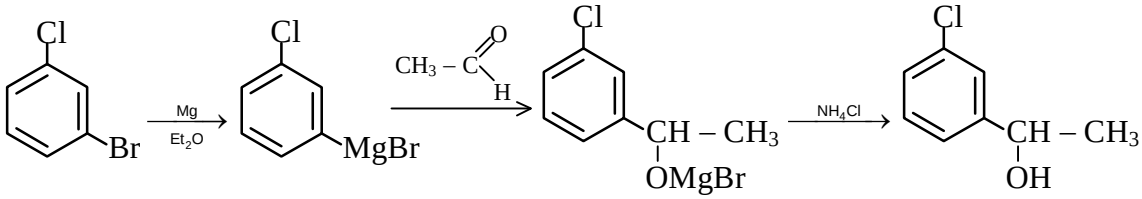
PART – II

SECTION – A

1. 

Hemiketal

2. Aryl chlorides do not form Grignard reagents in ether and THF is necessary as solvent, while aryl bromide and iodide readily react in ether solution.
The aqueous NH_4Cl is used to prevent the possibility of dehydration of alcohol which might occur if acid is used.



3. Let *x* ml ozone be present in 150 ml of ozonized oxygen.

$$2O_3 \xrightarrow[\text{tube}]{\text{heated}} 3O_2$$

x ml 3/2 *x* ml

Increasement in volume due to decomposition of ozone = 0.5 *x* = 10 ml

$\therefore$ *x* = 20 ml

Therefore, percentage of ozone in ozonized oxygen = $\frac{20}{150} \times 100 = 13.33\%$

4. Solubility of $PbSO_4 = \sqrt{K_{sp}} = \sqrt{1.44 \times 10^{-8}} = 1.2 \times 10^{-4}$ M

Solubility of $PbSO_4 = 1.2 \times 10^{-4} \times 303 \times 10^3 = 36.36$ mg litre^{-1}

$\therefore$ Volume of water needed to dissolve 1 mg of $PbSO_4 = \frac{1000}{36.36} = 27.5$ ml

5. The reaction is known as bromodesilylation.

6. Sulphurous acid is actually aqueous solution of SO_2

$$2NaOH + SO_2 \longrightarrow Na_2SO_3 + H_2O$$

8. Both C and CO act as reducing agent. Being gas CO reduction is achieved at the top of furnace at lower temperature of 250°C.

9. $[CuCl_5]^{3-}$ is $d_z sp^3$ hybridized and trigonal bipyramidal in shape while $[Ni(CN)_5]^{3-}$ is $d_{x^2-y^2} sp^3$ hybridized and square pyramidal in shape.

10. As chemisorption is monolayer phenomenon while physiosorption is multilayered adsorption.

SECTION – C

1. $\underset{(A)}{K_2Cr_2O_7} + H_2SO_4 + Cl^- \longrightarrow \underset{(B)}{CrO_2Cl_2} \xrightarrow{NaOH} \underset{(yellow)}{Na_2CrO_4} \xrightarrow{H^+} Na_2Cr_2O_7$

$\downarrow H_2O_2$

blue colour of CrO_5

2. $\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For $n_1 = 1$, $n_2 = ?$

$$\frac{1}{30.4\times10^{-9}} = 1.09678\times10^7 \times 4\left[\frac{1}{1^2} - \frac{1}{n_2^2}\right]$$

$\therefore$ **$n_2 = 2$**

For $n_1 = 2$ (first excited state), $n_2 = ?$

$$\frac{1}{108.5\times10^{-9}} = 1.09678\times10^7 \times 4\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$$

$\therefore$ **$n_2 = 5$**

3. $L = \sqrt{\ell(\ell+1)}\ \frac{h}{2\pi}$

as for 1 s orbital $\ell = 0$ $\therefore L = 0$

4. This octahedral complex involve d^2sp^3 hybridization which is hybrid of 6 atomic orbitals.

5. $pH = pK_a + \log\frac{[S]}{[A]}$

$pH = pK_a + \log 2$

now acid added so that $\frac{[S]}{[A]} = 1$

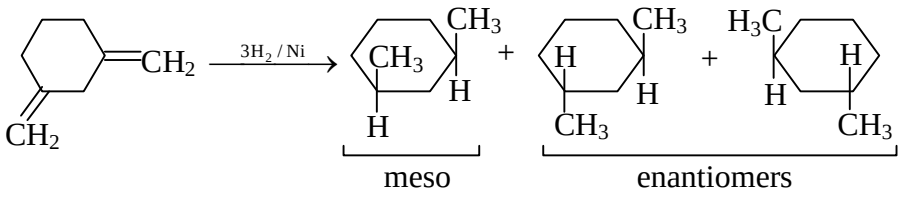
$pH' = pK_a + \log 1$

$pH' = pK_a$

$\therefore$ $\Delta pH = pK_a + \log 2 - pK_a$

$= \log 2 = 0.3$

$\therefore$ $10\ \Delta pH = 3$.

6. 

7. Let the 20 ml alkali solution → x m mole NaOH, y m mole Na_2CO_3

Using phenolphthalein indicator
nf of NaOH = 1
nf of Na_2CO_3 = 1
No. of meq. of base = $1 \times x + 1 \times y = x + y$
Meq. of acid = $60 \times 0.5 = 30$
Meq. of acid = Meq. base
$x + y = 30$
using methylorange indicator
No. of meq. of base $Na_2CO_3 = 1 \times y = 4$
Meq. of acid = $20 \times 0.5 = 10$
$y = 10$
$x = 20$
Molarity of NaOH = $\frac{20}{20} = 1$

8. $MnO_4^- \longrightarrow Mn^{2+}$ n- factor = 5

Equivalents of $MnO_4^- = 1.61\times10^{-3}\times5 = 8.05\times10^{-3}$

Equivalents of $A^{n+} = 8.05\times10^{-3}$

N –factor of $AO_3^- = 5 - n$ $\quad\therefore (5-n)\times2.68\times10^{-3} = 8.05\times10^{-3}$

$5 - n = 3$ $\quad n = 2$

9. $[\text{Pyridinium Chloride}] = \frac{0.15\times1000}{500} = 0.3M$ $\quad[\text{Pyridine}] = 0.2M$

A mixture of pyridine and its salt pyridinium chloride forms a basic buffer and therefore

$$pOH = -\log K_b + \log\frac{[\text{Salt}]}{[\text{Base}]}$$

$pOH = -\log\left(1.5\times10^{-9}\right) + \log\frac{0.30}{0.20}$ $\quad pOH = -\log1.5 - \log10^{-9} + \log1.5$

$pOH = 9$ $\quad pH = 14 - 9$ $\quad pH = 5$

10. No of mole of solute $\left(n_B\right) = \frac{6.2}{62} = 0.1$ $\quad\Delta Tf = Kf\times m$

$3.72 = 1.86\times\frac{0.1\times1000}{w_A}$ $\quad W_A = \frac{1.86\times0.1\times1000}{3.72}$

$W_A = 50$

Weight of ice separated from solution $= 55 - 50 = 5$ grams

Mathematics

PART – III

SECTION – A

1. Con-focal conics; therefore orthogonal intersection and circle touches at vertices

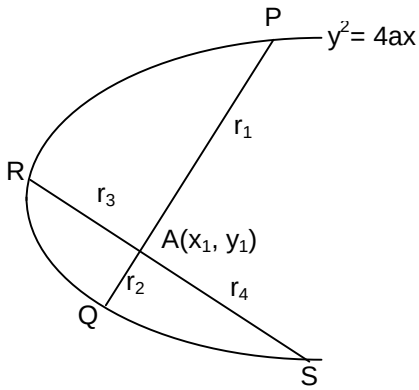
2. $\because f(x) - x^2 = (x-1)^2(x-3)$

3. $\sin|z| > 0$
$|z| \in (0,\pi)$ or $(2\pi, 3\pi)$ or $(4\pi, 5\pi)$
$\Rightarrow 0 < |z| < \pi,\ 2\pi < |z| < 3\pi$ so on

4. Let equation be $y = mx + c$ since it touches the hyperbola $c^2 = \frac{8}{9}(m^2-1)$
Also it touches the parabola $y^2 = 32x$, we get $c = \frac{8}{m}$
$\Rightarrow \frac{8}{9}(m^2-1) = \frac{64}{m^2} \Rightarrow (m^2-9)(m^2+8) = 0 \Rightarrow m \pm 3 \quad c = \pm\frac{8}{3}$
Requires equation are $9x - 3y + 8 = 0,\ 9x + 3y + 8 = 0$

5. $|adj(kA)| = |kA|^{n-1} = k^{n^2-n}|A|^{n-1}$ (A) is correct
(B) Order of C is unknown. Order of C not necessarily 3×3 hence B, C need not be equal
$\therefore$ (B) is incorrect
Since AB = O and A is non-singular
$\Rightarrow A^{-1}(AB) = A^{-1}O \Rightarrow IB = O \Rightarrow B = O$
(C) is correct
Now $|AB| \neq 0 \Rightarrow |A| \neq 0$ and $|B| \neq 0$
$\therefore$ A an B both are non-singular
$\Rightarrow$ (D) is correct

6. Let two perpendicular chords through $A(x_1, y_1)$ be PQ and RS



Equation of PQ is $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$
Where $\tan\theta$ = slope of PQ
Any point on this line may be taken as
$(x_1 + r\cos\theta, y_1 + r\sin\theta)$
As the point lies on $y^2 = 4ax$,
$\therefore (y_1 + r\sin\theta)^2 = 4a(x_1 + r\cos\theta)$
$\Rightarrow r^2\sin^2\theta + 2y_1\sin\theta . r - 4a\cos\theta r + y_1^2 - 4ax_1 = 0$

$\therefore r_1r_2 = \frac{(y_1^2 - 4ax_1)}{\sin^2\theta}$

Similarly $r_3r_4 = \frac{(y_1^2 - 4ax_1)}{\sin^2\alpha}$ where $\tan\alpha$ = slope of RS
Since PQ is perpendicular to RS,

$\therefore \alpha = 90 + \theta$ or $\theta - 90$

In either case; $\sin^2 \alpha = \cos^2 \theta$

Now $\frac{1}{r_1 r_2} + \frac{1}{r_3 r_4} = \frac{\sin^2\theta + \cos^2\theta}{\left(y_1^2 - 4ax_1\right)} = \frac{1}{\left(y_1^2 - 4ax_1\right)}$ which is constant

7. Since $\tan(5\pi \cos\theta) = \cot(5\pi \sin\theta)$

Or $\sin\theta + \cos\theta = \frac{2n+1}{10}$, $n \in I$

$\Rightarrow -1 < \frac{2n+1}{10\sqrt{2}} < 1 \Rightarrow -\left(\frac{10\sqrt{2}+1}{2}\right) < n < \left(\frac{10\sqrt{2}-1}{2}\right)$

Hence total number of solution is 14

8. $A^2 = -A \Rightarrow |A|^2 = -|A|$ if n is odd

$\Rightarrow |A| = -1$, $A^n \text{ adj } A^n = |A^n| I$

$\Rightarrow |A^n| \ |\text{Adj } A^n| = |A^n I|$

$\Rightarrow |\text{Adj } A^n| = 1$

Similarly when n is even $|A| = 1$

$\Rightarrow |\text{Adj } A^n| = 1$

9. $\log_3 \log_{27} \log_4 64 = -1$

$2\log_{18}\left(\sqrt{2} + \sqrt{8}\right) = \log_{\left(3\sqrt{2}\right)}\left(\sqrt{2} + 2\sqrt{2}\right) = \log_{3\sqrt{2}}\left(3\sqrt{2}\right) = 1$

$\log_2 \sqrt{10} + \log_2\left(\frac{2}{\sqrt{5}}\right) = \log_2\left(\sqrt{10} \times \frac{2}{\sqrt{5}}\right) = \log_2\left(2\sqrt{2}\right) = \frac{3}{2}$

$-\log_{\sqrt{2}-1}\left(\sqrt{2}+1\right) = 1$

10. $\int \frac{dx}{p^2 \sin^2 x + r^2 \cos^2 x} = \frac{1}{12}\tan^{-1}(3\tan x) + c$

$\int \frac{\sec^2 x}{p^2 \tan^2 x + r^2} dx$ ($p \tan x = t$, $p \sec^2 x \, dx = dt$)

$\frac{1}{p}\int \frac{dt}{t^2 + r^2} = \frac{1}{pr}\tan^{-1}\left(\frac{t}{r}\right) + c = \frac{1}{pr}\tan^{-1}\left(\frac{p\tan x}{r}\right) + c$

$\therefore pr = 12$, $\frac{p}{r} = 3 \Rightarrow p = 6, r = 2$

$\therefore -6 < 6\sin x + 2\cos x < 6$ when $\sin x = \pm 1$

SECTION – C

1. When $a = 2, x = 3$

2. Centres are (0, 4), (1, 0) and radii are each unity

$\Rightarrow$ The shortest distance = distance b/w centres – (sum of radii) $= \sqrt{17} - 2$

Maximum distance = distance b/w centres + (sum of radii) $= \sqrt{17} + 2$

$\therefore$ Diff. = 4

3. Let $\overrightarrow{OA} = x_1 i + x_1^2 j$, $\overrightarrow{OB} = x_2 i + x_2^2 j$

$\overrightarrow{OA} j = 1 \Rightarrow x_1 i \,.\, i + x_1^2 j \,.\, i = 1$

$\Rightarrow x_1 = 1 \quad \therefore y_1 = x_1^2 = 1$

$\overrightarrow{OB} \,.\, j = 4 = (x_2 i + x_2^2 j) \,.j = 4$

$\Rightarrow x_2^2 = 4 \Rightarrow x_2 = -2$ ($\because$ B lies in II quad.)

$\overrightarrow{OA}.\overrightarrow{OB} = (i + j).(-2i + 4j) = -2 + 4 = 2$

4. $175 = 5^2.7,\ 245 = 5.7^2,\ 875 = 5^3.7,\ 1715 = 5.7^3$

Let $\alpha = \log 5, \qquad \beta = \log 7$

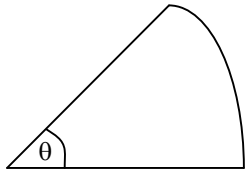
$$a = \frac{\log 175}{\log 245} = \frac{2\alpha + \beta}{\alpha + 2\beta}$$

$$b = \frac{\log 875}{\log 1715} = \frac{3\alpha + \beta}{\alpha + 3\beta}$$

$$\frac{1 - ab}{a - b} = \frac{(\alpha + 2\beta)\,(\alpha + 3\beta) - (2\alpha + \beta)\,(3\alpha + \beta)}{(2\alpha + \beta)\,(\alpha + 3\beta) - (\alpha + 2\beta)\,(3\alpha + \beta)}$$

$$= \frac{5(\beta^2 - \alpha^2)}{\beta^2 - \alpha^2} = 5$$

5. Let length $= \ell$ radius of sector $= r$, angle $= \theta$



$$\theta = \frac{\ell - 2r}{r}$$

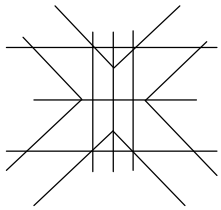
$$A = \frac{1}{2} r^2 . \theta = \frac{1}{2} r^2 \left(\frac{\ell - 2r}{r} \right) = \frac{1}{2} \left(\ell r - 2r^2 \right)$$

$$\frac{dA}{dr} = \frac{\ell - 4r}{2} = 0 \qquad r = \frac{\ell}{4}, \quad \frac{d^2 A}{dr^2} = -2 < 0$$

$$A_{max} = \frac{r}{2}(4r - 2r) = r^2 = \ell^2 / 16$$

Max. area of rectangle $= \left(\frac{\ell}{4} \right)^2 = \frac{\ell^2}{16}$

8. Shown in the figure



10. $\left[\frac{x}{9} \right] = \left[\frac{x}{11} \right] = I$

$\Rightarrow I \le \frac{x}{9} < I + 1, I \le \frac{x}{11} < I + 1,$ Solution is possible only when $11I < 9I + 9$

$\Rightarrow 0 \le I < \frac{9}{2}$

Total number of solution $\sum_{I=0}^{4} (9 - 2I) - 1 = 24$